FEM for 1D problems

Summary and discussion of results

Real world problem

Actual displacement

Computer model

① Strong form (D)
Differential equ. + BCs

②

Weak form (V)
$a(u, \delta u) = b(\delta u)$

Approximation
$u_h = \sum_{i=1}^{N} \varphi_i \cdot \hat{u}_i$

③

④

Discrete equation
$\underline{K}\,\underline{\hat{u}} = \underline{r}$

⑥

Coefficients of u_h
$\hat{u}_1, \hat{u}_2, \dots, \hat{u}_N$

⑤

Problem Data
Material, geometry, BCs

⑦

Predicted displacement

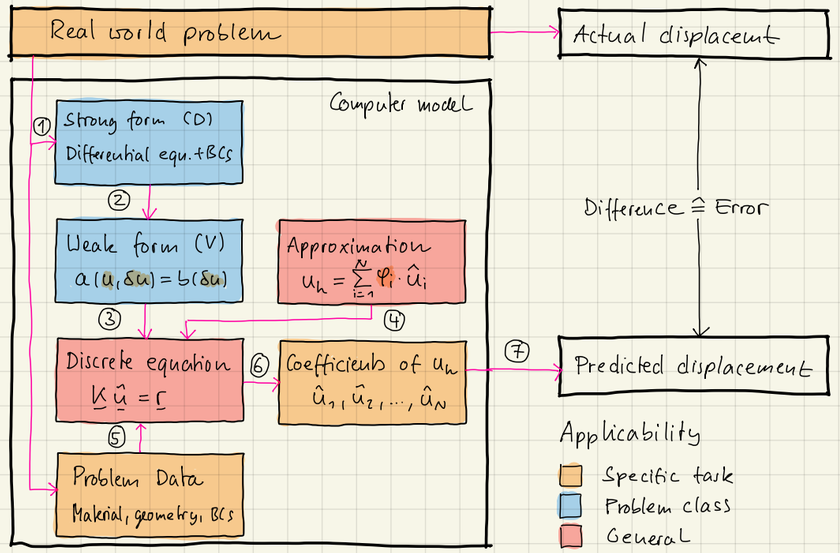
Difference ≙ Error

Applicability

- Specific task
- Problem class
- General

Errors introduced in each step

1. **Modelling error**
   - Nonlinear material behaviour of soil
   - Material properties vary
2. None here
3. None here
4. **Discretization error** $e = u - u_h$
5. **Data error** GIGO (Garbage in, garbage out)
6. Roundoff errors
   - Most of the time not a problem
7. **Interpration error**
   - Draw the right conclusions!



Validation and verification

Validation – Do the equations in (D) capture all relevant effects?

"Do I solve the right equation?"

Verification – Does my numerical solution solve (D)?

"Do I solve the equation right?"